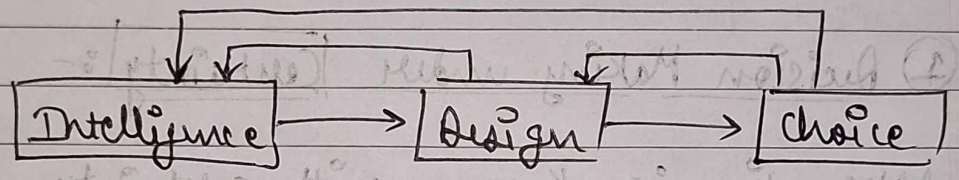
Decision Modeling
Week - 1

Decision Analysis

↳ analytical & systematic approach → optimal decision strategy → several decision alternatives



Decision Making Process
(IDC)

★

Rational/Normative decision makers	Administrative/Descriptive decision makers
① Identify problem, decision criteria	① Use incomplete and imperfect information
② Allocate weights to criteria	② Constrained by bounded rationality
③ Develop and analyze alternatives	③ Tend to satisfice immediate concerns
④ Select the "best" choice.	④ Choose a decision that may not be optimum but only satisficing.
⑤ Implement	
⑥ Evaluate "effectiveness"	
Structured ↑ Normative	Not structured ↑ Descriptive

(A) Decision Making environments

- Certainty
- Risk
- Uncertainty

Low ← Moderate → High

Ambiguity and chances of making a bad decision

① Decision Making under Certainty :-

Alternatives :- Known with certainty

Outcomes :- Known with certainty

Decision-making → Straightforward (no ambiguity)

Models → Linear Programming
→ Analytic Hierarchy Process
→ EOQ and EPQ models

② Decision Making under Risk :-

Alternatives & Outcomes → Known

Alternatives ↓ Outcomes → known only with probabilities associated (known)

Decision-making → At risk (ambiguities due to probabilities)

③ Decision Making under Uncertainty :-

Alternatives and Outcomes :- Unknown
(Not known with certainty)

Alternatives ↓ Outcomes → Not known → even probabilities are not known

Decision-Making → uncertain
(high ambiguity)

⊛ Criteria for Decision Making under Uncertainty

1. Maximax or Optimism criterion
2. Maximin or Pessimism criterion
3. Hurwicz or criterion of Realism
4. Laplace or Equally likely criterion
5. Savage or Minimax regret criterion

Expected Value or Expected Monetary Value (EMV)

1. Obtain probability of occurrence (P_i) for each state of nature
2. Find the expected value for each decision alternative as the sum of (P_i × Payoff i) for all the states of nature
3. Select the decision alternative with the best expected value

Decision Making under Risk

Expected Value of Perfect Information :

→ The gain in expected return obtained from knowing with certainty the future state of nature

→ EVPI = Max. amount a decision maker would pay for additional info

Bayesian Analysis :-

$$\boxed{\text{Prior Probabilities}} \longrightarrow \underbrace{\boxed{\text{New Data}} \longrightarrow}_{\text{Bayes Theorem}} \boxed{\text{Posterior Probabilities}}$$

Payoff Matrix :-

Decision Alternatives	States of Nature			Payoff for a Decision Strategy
	—	—	—	
	—	—	—	
	—	—	—	
	—	—	—	
	—	—	—	

Degree of optimism $\alpha = 0.4$

Payoff Matrix (Example)

Decision Alternatives				Maxi Max	Maxi Min	Hu ru iz
Redisgn	80	100	-80	100	-80	-8
Refurbish	50	200	-200	200	-200	-40
Do nothing	0	80	-50	50	-50	-18
States of Nature →	No Market changes	Favourable market changes	Unfavourable market changes			

States of nature → not under the control of decision maker

Decision alternatives → under the control of the decision maker

Regret Matrix

	(0.4) No Market changes	(0.4) Favourable market changes	(0.2) Unfavourable market changes	Expected Regret
Redisgn	20	100	30	54
Refurbish	0	0	150	30
Do nothing	50	170	0	88

Decision Savage Decision criteria

① Maximax : Optimism Criterion :-

→ Maximum possible payoff for each decision alternative

→ Select the decision alternative with the maximum of the above max. payoffs

② Maximin : Pessimism Criterion :-

→ Find the minimum possible payoff for each decision alternative

→ Select the decision alternative with the maximum of the above min. payoffs

③ Hurwicz : Criterion of Realism :-

→ Choose a degree of optimism (a). Obtain max. payoff and min. payoff for each decision alternative.

→ Hurwicz payoffs for each decision alternative are :- $a*(Max) + (1-a)*(Min)$

→ Select the decision alternative with the max. of above Hurwicz payoffs

④ Laplace : Equally likely criterion :-

→ Find the average value of payoff for each decision alternative

→ Select the decision alternative with the max. of the above average payoff values

⑤ Savage : Minimax Regret Criterion :-

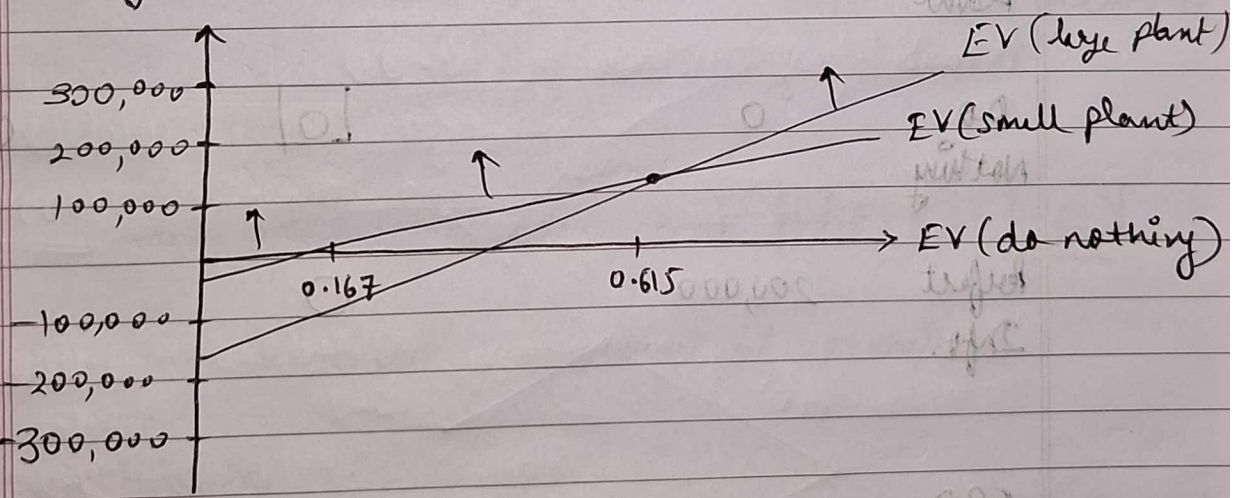
→ Find the regret matrix. (For each payoff under a state of nature, calculate difference from max payoff)

→ Obtain the max. regret for each decision alternative

→ Select the decision alternative with the min. of the above max. regrets.

Sensitivity Analysis :-

	(P) Favourable	(1-P) Unfavourable	Expected Value
Construct a large plant	200,000	-180,000	$= 200,000P - 180,000(1-P)$ $= 380,000P - 180,00$
Construct a small plant	100,000	-20,000	$= 100,000P - (20,000)(1-P)$ $= 120,000P - 20,000$
Do nothing	0	0	$0P + 0(1-P) = 0$



$P > 0.615$ → Large plant

$0.167 < P < 0.615$ → Small point

$P < 0.167$ → Do nothing

Expected Value of Perfect Information:-

$$\boxed{EVPI = ERPI - \text{best } EREV}$$

ERPI - expected return with perfect info

EREV - expected return with expected value approach

Alternative	States of Nature: Favourable Market (0.5)	Unfavourable Market (0.5)	(EREV) Expected Value (EV)
Large Plant	$\boxed{200,000}$	-180,000	10,000
Small Plant	100,000	-20,000	~~40,000~~ $\boxed{40,000}$
Do Nothing	0	$\boxed{0}$	0
Perfect Info.	200,000	0	

$$ERPI = 200,000 * 0.5 + 0 * 0.5$$

$$= \boxed{100,000}$$

$$EVPI = ERPI - \text{best } EREV = 100,000 - 40,000$$

Value of Information = $\boxed{\$60,000}$

Probability Concepts :-

Probability → The likelihood or the chance that an uncertain event would happen.

Mutually Exclusive Events :-

$P(E_1 \text{ or } E_2) = P(E_1 \cup E_2) = P(E_1) + P(E_2)$

Overlapping Events :-

$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

Conditional Probability :-

$\boxed{P(E|F) = \frac{P(E \cap F)}{P(F)}}$ = P(E occurs given that F has occurred)

Independent Events :-

$\boxed{P(E \cap F) = P(E)P(F)}$ | $\boxed{P(E|F) = P(E)}$

↳ E & F are independent events.

Imp Ques

EVPI is always equal to MER // (Min. expected regret)

$\boxed{EVPI = MER}$.